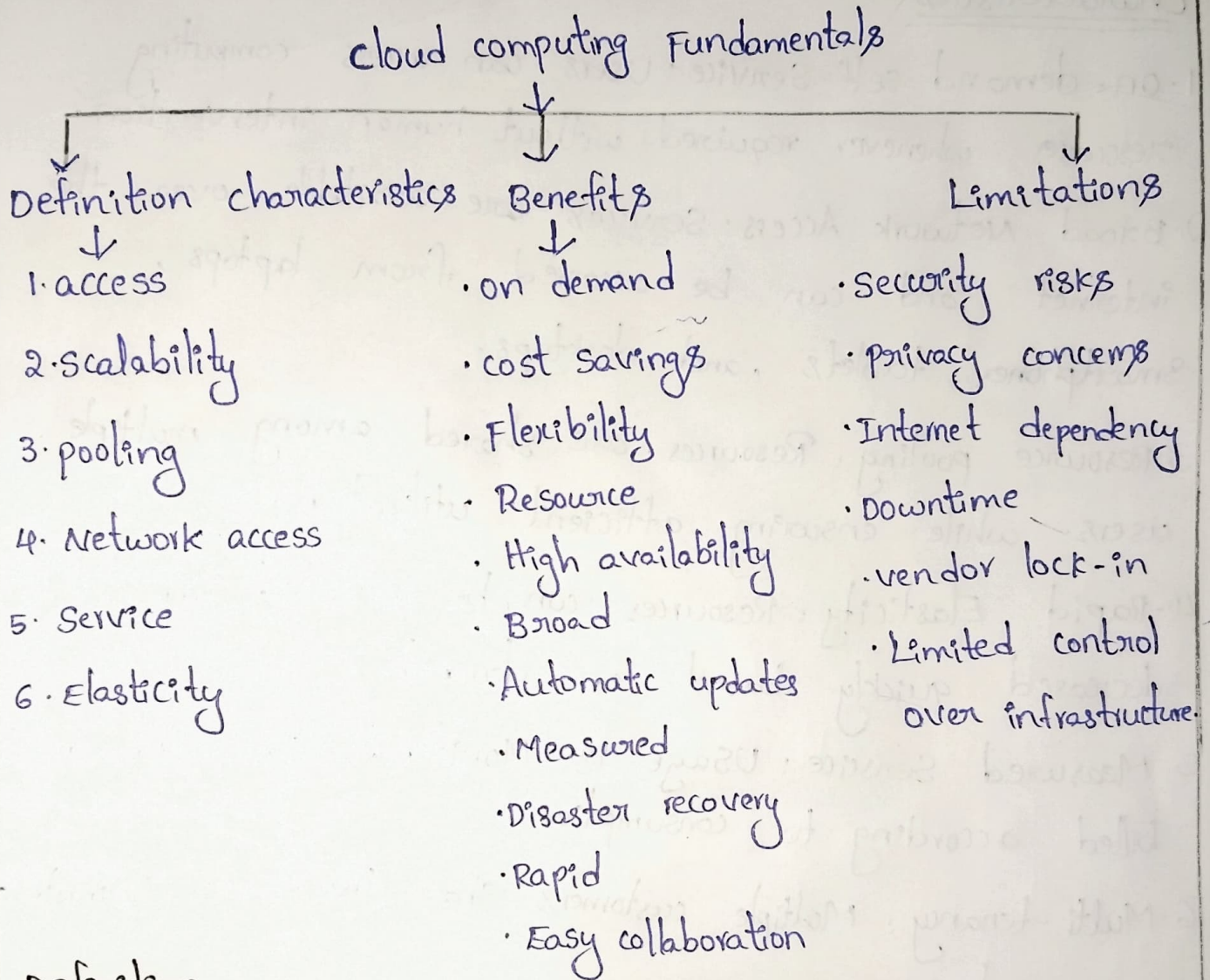


ASSESSMENT TASKS

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cloud computing

1. Concept Map 1: cloud computing Fundamentals. Include definition, characteristics, benefits, and limitations.



Definition

cloud computing is a technology that delivers computing resources such as servers, storage, databases, networking, software, and analytics through the internet. Instead of purchasing and maintaining physical hardware, users can access these services whenever needed. cloud

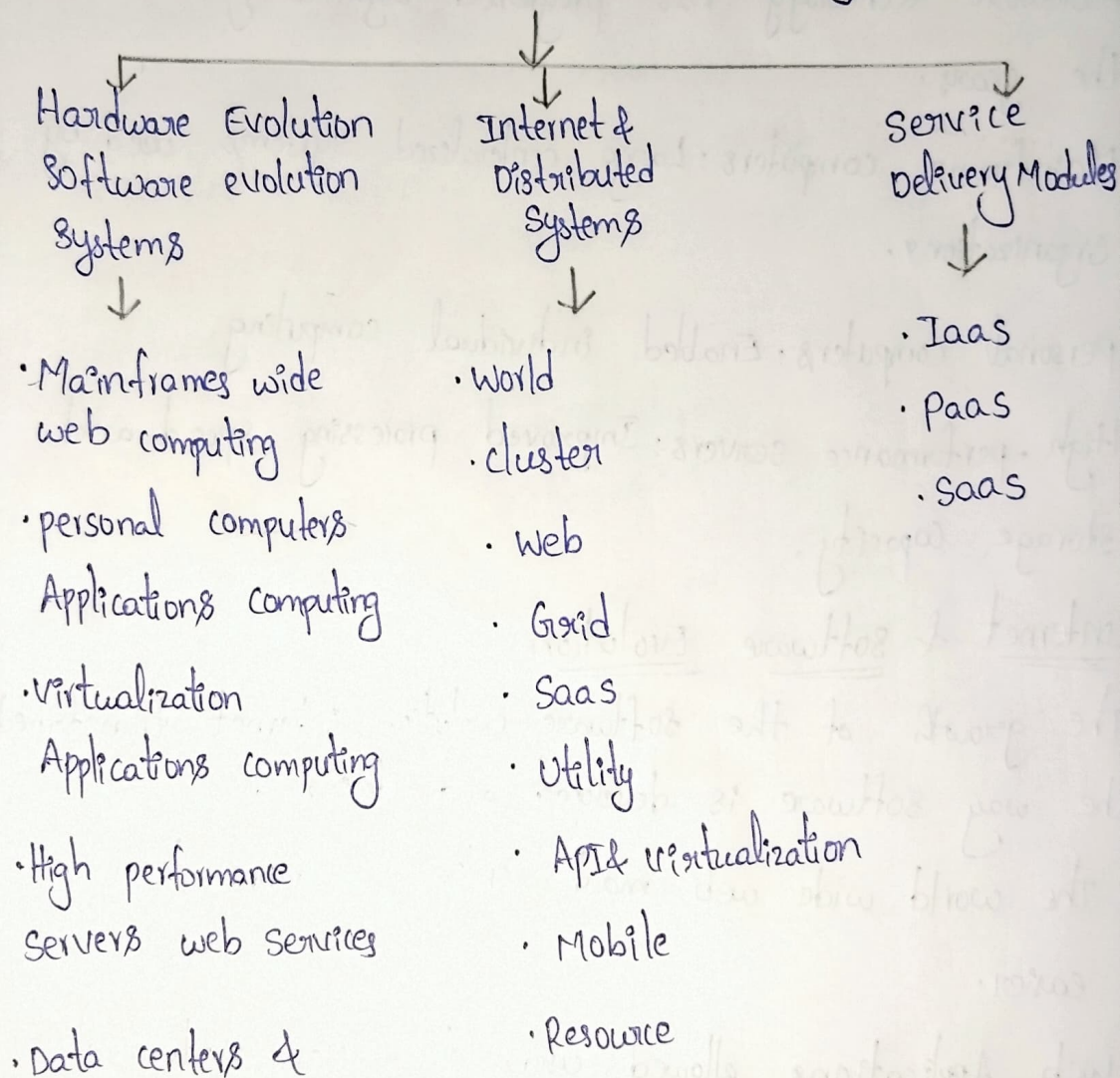
Computing allows individuals and organizations to store data, run applications, and manage workloads remotely with a pay-as-you-go pricing model.

Characteristics

1. On-demand self-Service: Users can access computing resources whenever required without human intervention.
2. Broad Network Access: Services are available over the internet and can be accessed from laptops, smartphones, tablets, and desktops.
3. Resource pooling: Resources are shared among multiple users while ensuring efficient utilization.
4. Rapid Elasticity: Resources can be increased or decreased quickly based on demand.
5. Measured Service: Usage is Monitored, controlled, and billed according to consumption.
6. Multi-tenancy: Multiple customers share the same infrastructure securely.
7. Pay-as-you-go: Users pay only for the resources they use.

2. Concept Map 2: Evolution of cloud computing. connect hardware evolution, internet software evolution, distributed systems, and service delivery.

Evolution of cloud computing



Evolution of cloud computing

cloud computing evolved through continuous improvements in computer hardware, networking technologies, software development, and distributed computing systems.

These developments made it possible to provide computing resources as online services.

Hardware Evolution

Hardware technology has progressed significantly over the years.

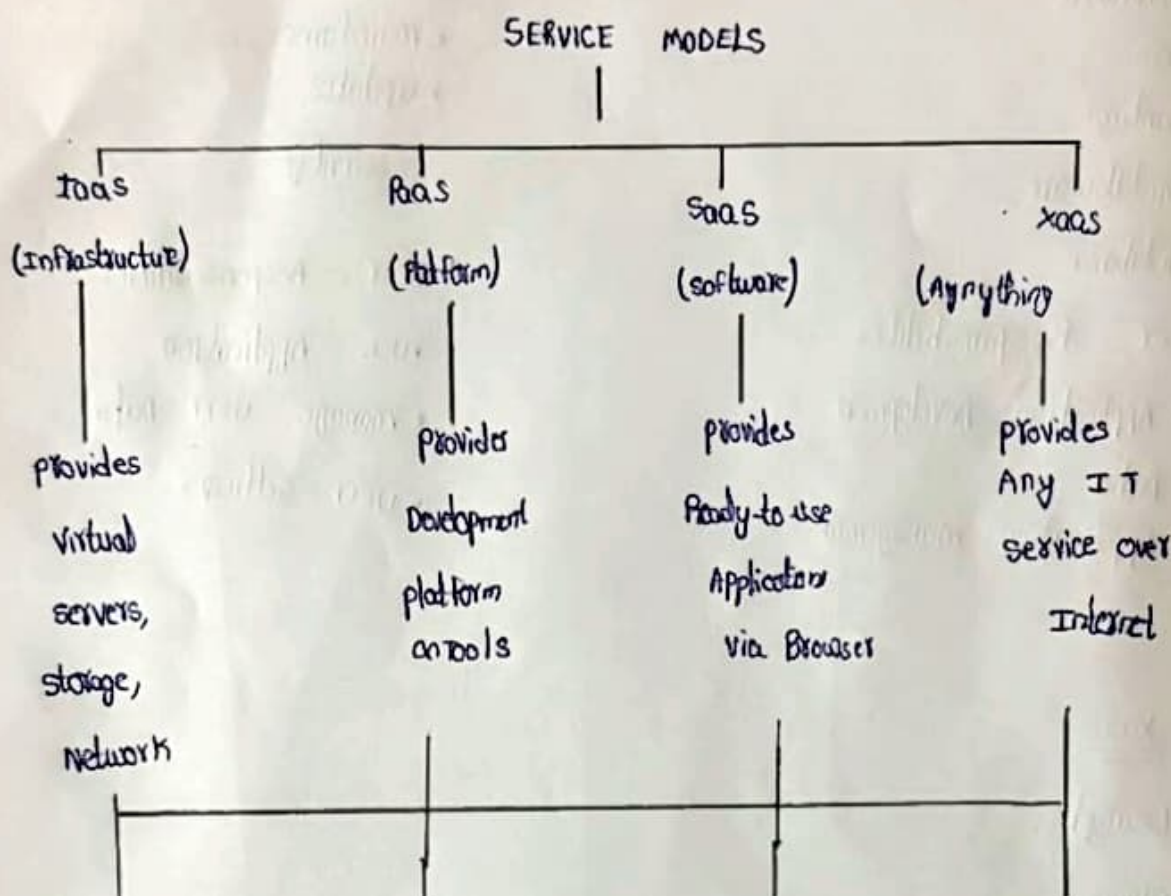
- Mainframe computers: Large centralized systems used by organizations.
- Personal computers: Enabled individual computing.
- High-performance servers: Improved processing speed and storage capacity.

Internet & Software Evolution

The growth of the software evolution internet transformed the way software is developed and delivered.

- The world wide web made online communication easier.
- Web Applications allowed users to access software through browsers.
- APIs and web services enabled communication between different applications.

3. concept map: service models IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities



IaaS
Provider Responsibilities

- Hardware
- storage
- Networking
- security of infrastructure

User Responsibilities

- install os
- Applications
- data
- security configuration

PaaS

Provider Responsibilities

- Hardware
- OS
- Runtime
- Middleware
- Database

User Responsibilities

- Application Development
- Data
- Application management

SaaS

Provider Responsibilities

- complete software
- maintenance
- updates
- security

User Responsibilities

- use Application
- manage user data
- user settings

XaaS

Examples:

- DaaS
- FaaS
- NaaS
- SaaS
- security as a service

4. concept map4: virtualization and web services as Enablers of cloud . show how these two concepts support cloud deployment and scalability.

